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Computer Programme (Literary Work) — Copyright Act, 1957

1. Particulars of the work

Particular	Details
Title of the work	Satya-Score: An AI Hallucination Detection & Scoring Framework
Nature of the work	Computer programme (software) with its documentation — literary work under Section 2(o) of the Copyright Act, 1957
Languages of the programme	Python 3 (back end, models, evaluation); TypeScript / React (web application)
Author(s)	Vivek Yeshwantrao Borkar, Piyush Devidas Nande, Aasawari Sahebrao Jumade, Arunima Tridip Wankhade
Project guide	Dr. R. R. Karwa
Applicant / owner	Prof. Ram Meghe Institute of Technology & Research, Badnera, Amravati (filing through the institute)
Department	Department of Artificial Intelligence and Data Science
Year of creation	2026
Publication status	Unpublished (source code held in a private repository)
Size of the original work	≈ 8,225 lines in the running system (5,817 Python, 2,408 TypeScript/TSX), plus 2,878 lines of automated tests, 7,016 lines of evaluation and data-preparation scripts and 437 lines of deployment scripts

2. Description of the work

Satya-Score checks text written by AI chatbots (large language models) for factual errors, known as hallucinations. A user pastes an answer in English or Hindi; the software splits it into individual factual claims, checks each claim against trusted sources, and marks every claim Verified, Uncertain or Hallucinated in place in the original text. Each verdict links to its evidence and, when Wikidata or official statistics decided it, states the correct fact (for example: “Correct fact: Mahatma Gandhi · place of birth: Porbandar”). A verdict for the whole text and a graded score are also given.

Each claim is checked against the most authoritative source that covers it: official statistics (World Bank Open Data and the WHO Global Health Observatory) and structured facts in the Wikidata knowledge graph first, then professional fact-checks (Google Fact Check Tools), and otherwise passages retrieved from English and Hindi Wikipedia judged by a natural-language-inference model. The software is delivered as a web application with user accounts, a history of checks, file upload, export and a documented programming interface (API).

3. Main functions

- **Claim extraction:** splits English text into self-contained claims (dependency parsing, pronoun resolution that follows the sentence topic, list handling) and Hindi text into sentences; skips questions, pleasantries, advice and opinions.

- **Evidence retrieval:** cross-lingual sentence embeddings and a FAISS index over 102,069 Wikipedia passages (English and Hindi), extended on demand with articles found by live Wikipedia search.
- **Evidence rule:** a natural-language-inference judgement per passage, combined so that support decides first and only a passage about the claim's own subject may contradict it.
- **Structured verification:** Wikidata checks for places, capitals, dates of birth, death and founding, authorship, current office-holders, heights, lengths, populations, areas and currencies, in English and Hindi; World Bank and WHO statistics with two-agency agreement; professional fact-checks applied only when logically entailed (negation-safe).
- **Scoring and explanation:** per-claim score set by the deciding source (statistics or Wikidata, then a fact-check, then the evidence rule) within the CMHSA scoring scheme, verdict thresholds, whole-text verdict by the share of hallucinated claims, overall score, the deciding source, and the supporting sentence or the correct fact.
- **Web application and API:** highlighted results, history, export (CSV/JSON), .txt/.pdf upload, accounts with scrypt-hashed passwords, password change and administrator-issued reset codes, CSRF protection, rate limits; usable on phones over a local network and from anywhere through a secure public link.

4. Structure of the programme

Module (path)	Function
member3_backend/claim_extractor.py	Claim extraction, pronoun resolution, filtering of non-factual sentences
member3_backend/pipeline.py	Orchestration of retrieval, judgement, structured checks, scoring, explanations; parallel lookups
member3_backend/evidence_rules.py	Evidence aggregation rule and source attribution
member3_backend/wiki_expander.py	On-demand Wikipedia expansion of the evidence index
member3_backend/factcheck_rules.py	Negation-safe matching of professional fact-checks
member3_backend/api.py, auth.py	Web API, accounts, sessions, history, security
member2_data/entity_linker.py, wikidata_connector.py	Entity linking (English, Hindi templates) and Wikidata verification
member2_data/kg_extra.py, dates.py	Authorship, office, quantity, currency and date checks
member2_data/official_stats.py, factcheck.py	World Bank / WHO statistics; fact-check search client
member1_model/*	Embeddings, NLI scorer, CMHSA scoring and weight training, calibration
web/src/*	Web application (React, TypeScript)
tests/*; *_eval.py	265 automated tests; evaluation scripts with pre-registered decision rules

5. Original contribution of the authors

The source code, the decision rules, the evaluation scripts, the web application and the documentation listed above were written by the authors. In particular the following are the authors' own design and expression:

- the routing of each claim to the most authoritative source and the explanation shown with each verdict;

- the negation-safe rule for applying a professional fact-check (a “False” rating applies only if the claim entails the checked claim; a “True” rating only if the checked claim entails the claim; never on a contradiction);
- the two-agency agreement rule for official statistics, with comparison at the precision a figure is written to;
- Hindi relation templates and exact-name linking with the namesake rule (any match supports; a contradiction needs every namesake to disagree);
- the evidence rule (support first; only an on-topic passage may contradict) and the topic-following pronoun resolution;
- the evaluation methodology: changes tested against decision rules fixed and recorded before the run, with negative results kept.

6. Third-party components (not claimed)

The following are used under their own licences and are not part of the work claimed: pretrained models (mDeBERTa-v3-base-mnli-xnli, MIT; paraphrase-xlm-r-multilingual-v1, Apache-2.0; spaCy en_core_web_sm, MIT); libraries (PyTorch, transformers, sentence-transformers, FAISS, FastAPI, spaCy, scikit-learn, pandas, React, Vite, Tailwind CSS and others); and data and services queried at run time (Wikipedia, CC BY-SA 4.0; Wikidata, CC0; World Bank Open Data, CC BY 4.0; WHO Global Health Observatory, CC BY-NC-SA 3.0 IGO, non-commercial; Google Fact Check Tools API). The full list is in THIRD_PARTY_NOTICES.md in the source code.

7. Source code to be attached

As required by Rule 70(5) of the Copyright Rules, 2013 (as amended in 2021), the first 10 and the last 10 pages of the source code are to be attached to the application, without redaction.

8. Declaration

We declare that the work described above is our original work, created during our project at Prof. Ram Meghe Institute of Technology & Research, Badnera, Amravati, and that the third-party components listed in Section 6 are not claimed.

Signatures of the authors:

Vivek Yeshwantrao Borkar: _____ Date: _____

Piyush Devidas Nande: _____ Date: _____

Aasawari Sahebrao Jumade: _____ Date: _____

Arunima Tridip Wankhade: _____ Date: _____

Project guide: _____ Date: _____